



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Algebraic Fractions

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Simplifying Algebraic Fractions

Simplifying Algebraic Fractions

What is an algebraic fraction?

- An **algebraic fraction** is a fraction with an **algebraic expression** on the top (numerator) and/or the bottom (denominator)

How do you simplify an algebraic fraction?

- If possible, **factorise fully** the top and bottom

- E.g. $\frac{2x}{x^2 + 3x} = \frac{2x}{x(x + 3)}$

- Cancel** common factors

- This factor may be a **single term**

- E.g. $\frac{\cancel{x}(5x - 1)}{4\cancel{x}} = \frac{5x - 1}{4}$

- It could also be a **common bracket**

- E.g. $\frac{x\cancel{(x + 2)}}{\cancel{(x + 2)}(x - 1)} = \frac{x}{x - 1}$

- A **common mistake** is to cancel a factor that is not common to **all terms** in either the top or the bottom of a fraction

- E.g. The fraction $\frac{6x}{x + 1}$ cannot be simplified

- x is not common to **all** terms in the bottom of the fraction
- and the expression on the bottom cannot be factorised



Examiner Tips and Tricks

- When asked to **simplify** an algebraic fraction, **factorise** top and bottom



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Worked Example

Simplify $\frac{4x + 6}{2x^2 - 7x - 15}$

Factorise the top, by using 2 as a common factor

$$\frac{2(2x + 3)}{2x^2 - 7x - 15}$$

Factorise the bottom using your preferred method

Using the fact that the top factorised to $(2x + 3)$ may help!

$$\frac{2(2x + 3)}{(2x + 3)(x - 5)}$$

The common factors on the top and bottom reduce to 1 (cancel out)

$$\frac{\cancel{2(2x + 3)}}{\cancel{(2x + 3)}(x - 5)}$$

$$\frac{2}{(x - 5)}$$

Test yourself [Next topic](#)



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Adding & Subtracting Algebraic Fractions

Adding & Subtracting Algebraic Fractions

How do I add (or subtract) two algebraic fractions?

- The **rules** for adding and subtracting **algebraic fractions** are the **same** as they are for **fractions with numbers**

STEP 1

Find the **lowest common denominator** (LCD)

- Sometimes the LCD can be found by **multiplying** the denominators together

- E.g. The LCD for the fractions $\frac{1}{x+2}$ and $\frac{1}{x+5}$ is $(x+2)(x+5)$

- Similarly, with numbers, the LCD of $\frac{1}{2}$ and $\frac{1}{5}$ is $2 \times 5 = 10$

- Although multiplying the denominators will always give you a multiple, it is **not necessarily the lowest** multiple

- E.g. The LCD for the fractions $\frac{1}{x}$ and $\frac{1}{2x}$ is $2x$ (not $2x^2$) as both terms already include an x

- Similarly, with numbers, the LCD of $\frac{1}{2}$ and $\frac{1}{4}$ is just 4, not $2 \times 4 = 8$

- Other examples include:

- The LCD of $\frac{1}{x+2}$ and $\frac{1}{(x+2)(x-1)}$ is $(x+2)(x-1)$

- The LCD of $\frac{1}{x+1}$ and $\frac{1}{(x+1)^2}$ is $(x+1)^2$

- The LCD of $\frac{1}{(x+3)(x-1)}$ and $\frac{1}{(x+4)(x-1)}$ is $(x+3)(x-1)(x+4)$

STEP 2

Write each fraction over the **lowest common denominator**



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Multiply the numerator of each fraction by the **same amount** as the denominator

$$\text{E.g. } \frac{x}{x-4} + \frac{1}{x+2} = \frac{x(x+2)}{(x-4)(x+2)} + \frac{(x-4)}{(x-4)(x+2)}$$

STEP 3

Write as a **single fraction** over the lowest common denominator and **simplify the numerator**

- Do this by **adding** or **subtracting** the **numerators**
- Take particular care if subtracting

$$\text{E.g. } \frac{x(x+2) + (x-4)}{(x-4)(x+2)} = \frac{x^2 + 2x + x - 4}{(x-4)(x+2)} = \frac{x^2 + 3x - 4}{(x-4)(x+2)}$$

STEP 4

Check at the end to see if the top **factorises** and the fraction can be **simplified**

- E.g. $\frac{(x+4)(x-1)}{(x-4)(x+2)}$, the top factorises but there are no common factors so it is in its most simple form



Examiner Tips and Tricks

- Leaving the top and bottom of your answer in **factorised form** will help you see if anything cancels at the end



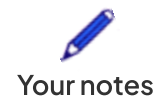
Worked Example

(a) Express $\frac{x}{x+4} - \frac{3}{x-1}$ as a single fraction.

The lowest common denominator is $(x+4)(x-1)$

Write each fraction over this common denominator, remember to multiply the top of the fractions too

$$\frac{x(x-1)}{(x+4)(x-1)} - \frac{3(x+4)}{(x-1)(x+4)}$$



Combine the fractions, as they now have the same denominator

$$\frac{x(x-1) - 3(x+4)}{(x+4)(x-1)}$$

Simplify the numerator

Be careful expanding with the negative signs

$$\frac{x^2 - x - 3x - 12}{(x+4)(x-1)} = \frac{x^2 - 4x - 12}{(x+4)(x-1)}$$

Factorise the top

$$\frac{(x+2)(x-6)}{(x+4)(x-1)}$$

There are no terms which would cancel here, so this is the final answer

$$\frac{(x+2)(x-6)}{(x+4)(x-1)}$$

(b) Express $\frac{x-4}{2(x-3)} - \frac{x-1}{2x}$ as a single fraction.

The lowest common denominator is $2x(x-3)$

(You could also use $4x(x-3)$ but this wouldn't be the lowest common denominator)

Write each fraction over this common denominator, remember to multiply the top of the fractions too

$$\frac{x(x-4)}{2x(x-3)} - \frac{(x-1)(x-3)}{2x(x-3)}$$

Combine the fractions, as they now have the same denominator

$$\frac{x(x-4) - (x-1)(x-3)}{2x(x-3)}$$

Simplify the numerator

Be careful expanding with negative signs

$$\frac{(x^2 - 4x) - (x^2 - 4x + 3)}{2x(x-3)} = \frac{x^2 - 4x - x^2 + 4x - 3}{2x(x-3)} = \frac{-3}{2x(x-3)}$$

There is nothing else that can be factorised on the numerator, so this is the final answer

$$\frac{-3}{2x(x-3)}$$

There are other accepted answers, e.g. $\frac{3}{2x(3-x)}$



Your notes



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Multiplying & Dividing Algebraic Fractions

Multiplying & Dividing Algebraic Fractions

How do I multiply algebraic fractions?

STEP 1

Simplify both fractions first by **fully factorising**

$$\text{E.g. } \frac{x}{3x+6} \times \frac{2x+4}{x+7} = \frac{x}{3(x+2)} \times \frac{2(x+2)}{x+7}$$

STEP 2

Cancel any common factors on top and bottom (from either fraction)

$$\text{E.g. } \frac{x}{\cancel{3(x+2)}} \times \frac{\cancel{2(x+2)}}{x+7} = \frac{x}{3} \times \frac{2}{x+7}$$

STEP 3

Multiply the **tops** together

Multiply the **bottoms** together

$$\text{E.g. } \frac{2x}{3(x+7)}$$

STEP 4

Check for any further factorising and cancelling

$$\text{E.g. } \frac{2x}{3(x+7)} \text{ has no common factors so is in its simplest form}$$

How do I divide algebraic fractions?

- Flip** (find the reciprocal of) the **second** fraction and replace $\div$ with $\times$

$$\text{So } \div \frac{a}{b} \text{ becomes } \times \frac{b}{a}$$

$$\text{E.g. } \frac{3x-12}{x} \div \frac{2x+8}{x+3} = \frac{3x-12}{x} \times \frac{x+3}{2x+8}$$

- Then follow the same rules for **multiplying** two fractions



Worked Example

Divide $\frac{x+3}{x-4}$ by $\frac{2x+6}{x^2-16}$, giving your answer as a simplified fraction.

Division is the same as multiplying by the reciprocal (the fraction flipped)

$$\frac{x+3}{x-4} \div \frac{2x+6}{x^2-16} = \frac{x+3}{x-4} \times \frac{x^2-16}{2x+6}$$

Factorise all numerators and denominators to see which factors cancel out

You need to use the difference of two squares, $x^2 - 4^2 = (x-4)(x+4)$

$$\frac{x+3}{x-4} \times \frac{x^2-16}{2x+6} = \frac{\cancel{x+3}}{\cancel{x-4}} \times \frac{(\cancel{x-4})(x+4)}{2(\cancel{x+3})}$$

Multiply the remaining numerators and denominators together

$$\frac{1 \times (x+4)}{1 \times 2} = \frac{x+4}{2}$$

Check to see if you missed any terms that are the same on the top and bottom that could be cancelled

$\frac{x+4}{2}$ is already in its simplest form

$$\frac{x+4}{2}$$



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Solving Equations with Algebraic Fractions

Solving Algebraic Fractions

How do I solve an equation that contains algebraic fractions?

- There are **two methods** for **solving equations** that contain algebraic fractions
- One method is to **add** or **subtract** the **algebraic fractions first** and then solve as usual

- For example, to solve $\frac{8}{x+1} - \frac{5}{x+2} = 1$

- First subtract the fractions and simplify, $\frac{3x+11}{(x+1)(x+2)} = 1$

- Then cross-multiply, expand and solve

$$3x + 11 = 1(x+1)(x+2)$$

$$3x + 11 = x^2 + 3x + 2$$

$$0 = x^2 - 9$$

$$0 = (x-3)(x+3)$$

$$x = 3 \text{ or } x = -3$$

- Alternatively, you can **remove the fractions first** by multiplying everything **on both sides** of the equation by each expression in the denominators and then solve

- For example, to solve the equation $\frac{4}{x-3} + \frac{5}{x+1} = 5$

- First multiply every term in the equation by both $(x-3)$ and $(x+1)$ and cancel common factors where possible

- Multiply every term by $(x-3)$ (this bracket goes in the **numerator** of any fractions)

$$\frac{4}{\cancel{(x-3)}} \cancel{(x-3)} + \frac{5(x-3)}{x+1} = 5(x-3)$$

$$4 + \frac{5(x-3)}{x+1} = 5(x-3)$$

- Then multiply every term by $(x+1)$



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$$4(x+1) + \frac{5(x-3)}{(x+1)}(x+1) = 5(x-3)(x+1)$$

$$4(x+1) + 5(x-3) = 5(x-3)(x+1)$$

- Then solve

$$4x + 4 + 5x - 15 = 5(x^2 - 2x - 3)$$

$$9x - 11 = 5x^2 - 10x - 15$$

$$0 = 5x^2 - 19x - 4$$

$$0 = (5x + 1)(x - 4)$$

$$x = -\frac{1}{5} \text{ or } x = 4$$



Examiner Tips and Tricks

- When multiplying by an algebraic expression, use **brackets** around the expression, e.g. $(2x + 3)$
- Multiplying by both denominators at once can speed up the process, but take care if choosing this technique in the exam!
 - and remember to multiply **all terms** on either side of the equation



Worked Example

$$\frac{2}{p+3} - \frac{5}{p} = 6p$$

Show that this equation can be written as $6p^3 + 18p^2 + 3p + 15 = 0$.

To clear the fractions, we multiply both sides of the equation by each denominator

Start by multiplying all terms in the equation by the denominator $(p + 3)$

The $(p + 3)$ on top and bottom will cancel in the first term

$$2 - \frac{5(p+3)}{p} = 6p(p+3)$$



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Now multiply all terms on both sides by the next denominator, p

The p on top and bottom will cancel in the second term

$$2(p) - 5(p + 3) = 6p(p + 3)(p)$$

Expand brackets

Be careful with negative signs

$$2p - 5(p + 3) = 6p^2(p + 3)$$

$$2p - 5p - 15 = 6p^3 + 18p^2$$

Collect like terms

$$-3p - 15 = 6p^3 + 18p^2$$

Add $3p$ and 15 to both sides of the equation

$$0 = 6p^3 + 18p^2 + 3p + 15$$

$$6p^3 + 18p^2 + 3p + 15 = 0$$